Indus Civilization as a punctuated equilibrium in social evolution

Introduction

The emergence of Indus Civilization has been studied with great interest, especially in relation to and in contrast with it's famous and prodigal neighbors, the great Sumerian and the Elamite civilizations. The sophisticated Indus civilization on the other hand occupies it's own quiet little mysterious corner. The emergence of civilization is generally understood as a process of continuous incremental development as in the Darwinian biological evolution. However, the biological evolution has discrete stages like the emergence of a biological species (Gould, 2007), a discrete stage in the continuous evolution, a quantum jump in a classical Newtonian process. Mathematically such a process has been popularized by Thom (1971), termed as catastrophe theory or more mundanely as bifurcation process. In condensed matter physics, a similar phenomena has been used to explained the sudden emergence of superconductivity in materials by Anderson (1972). We propose a similar process, a punctuated equilibrium, a new stage termed as "civilization", a phenomena wherein multiple professions come together to create a drastically new urban society in the Bronze age. Here we first reformulate the Sumerian civilization in this new framework before we go on to describe the Indus Civilization.

Urban Mesopotamian civilization: a paradigm shift

Although Northern regions of Mesopotamia developed earlier, it was the Southern regions that blossomed in urban culture is scale and sophistication. It was the alluvial plains of the south that became the vanguards of the civilization.

It was Jacobsen (1960) who first articulated that irrigation system was central to the Sumerian civilization, an important life-line and nervous system of the civilization that not only sustained farming and the urban life, but also become a connecting line to the river transport and trade. Oates (1960) cautioned that we need to consider littoral resources of the marsh lands like fish. Pournell and Algaze (2015) have argued although Jacobsen emphasis on irrigation system is primarily correct but at the same time the simultaneous coming together of three resources: (1) alluvial plains; (2) pasture lands; (3) marsh lands, is also important. It must be further emphasized that the creation of an organized canal system also enables the coming together of multiple resources, creating the greater urban society of Mesopotamia. The simultaneous coming together of many societies with their resources into greater conglomeration is what we call the urban civilization, an emergent phenomena. This new stable equilibrium is what we call the *punctuated equilibrium*.

One of the by-products of such a society is the emergence of a writing system. Postgate et al. (1995) studied the emergence of writing systems considering the two opposing view-points, one being that it is part of religious ritualist tradition and the other being that it arises as a utilitarian system, especially in relation to trade activities. Postgate et al. (1995) show clearly that administration and trade are the dominating factors in the impetus to create a writing system although ritualistic conventions are also a factor. We extend this hypothesis further stating that even the ritualistic traditions also arise from utilitarian traditions of an earlier society, now reduced in importance and stature. For example, the goddess Enki arises from the earlier society of folk tradition, a kind of Mermaid of the sea that sometimes swallows it's fisherman in the sea due to calamities in the sea. Sudden tragedy of a Tsunami may also be embedded in their great respect for Enki, the sea goddess, a remnant of the matriarchal society. As the urban civilization develops rapidly these remnants remain as ritualistic traditions without any utilitarian purpose and is gradually reduced in importance. Sometimes a strong ruling class uses these outmoded (non-utilitarian) traditions as their class identity. It then creates an ambiance that produces direct military conflict between different ritualistic traditions in the neighboring cultures. However, this not always the case, as a society does exist that has developed without those ritualistic identities, their temples and kings, namely, the Indus Valley. We shall now delve into this Indus society.

Emergence of Urban Civilizations in Indus Valley

Just as in Mesopotomia, the earliest developments in the Indus Valley Civilization happened in the Northern hilly regions of Mehgarh (meaning the abode of mountains), and not in the alluvial plains. Most of knowledge about the Indus script we use here is derived from the author's decipherment of Indus script (Venkatesan, 2025). Of course, we also derive a lot from the archaeological findings (Possehl, 1996; Kenoyer, 2006; Vidale, 2026). In this short description we follow the stages of development of the Indus script during the urban phase, i.e., the development of towns (ūr).

Phase 1

We can note the possible creation of logo-syllables like trident (vēl), first/chief (mutal), double (iru), unite/five (ai), fertile/crop (viri), plough (ēr), cross-way (cēri).

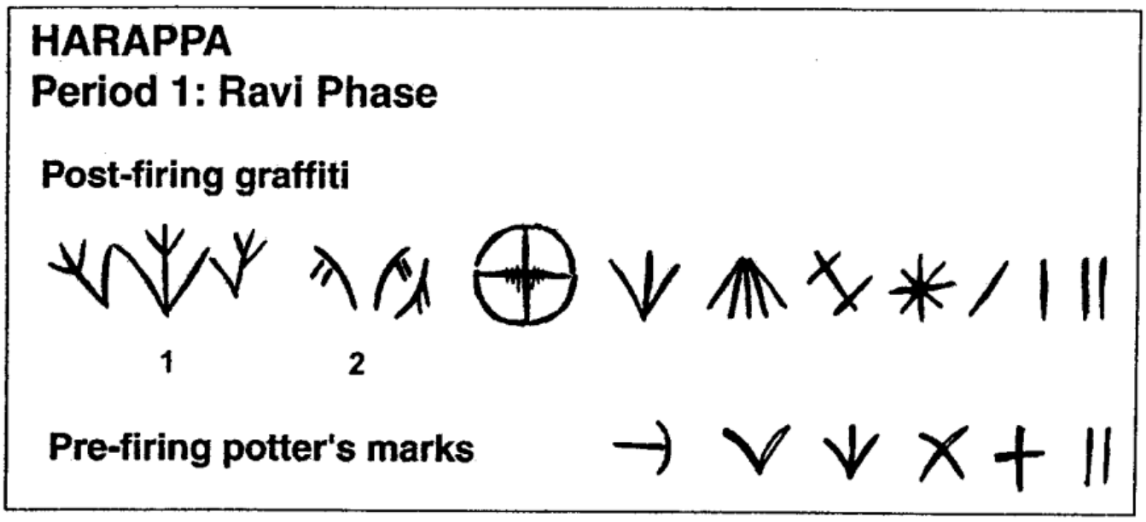

Figure 3. Harappa, Ravi Phase Script

The sun is star-shaped ✳.
The above Potter's marks were observed in the first phase, the Ravi Phase (3200-2800 BCE).
The tools they used to make the mark were bone tools:

Phase 2

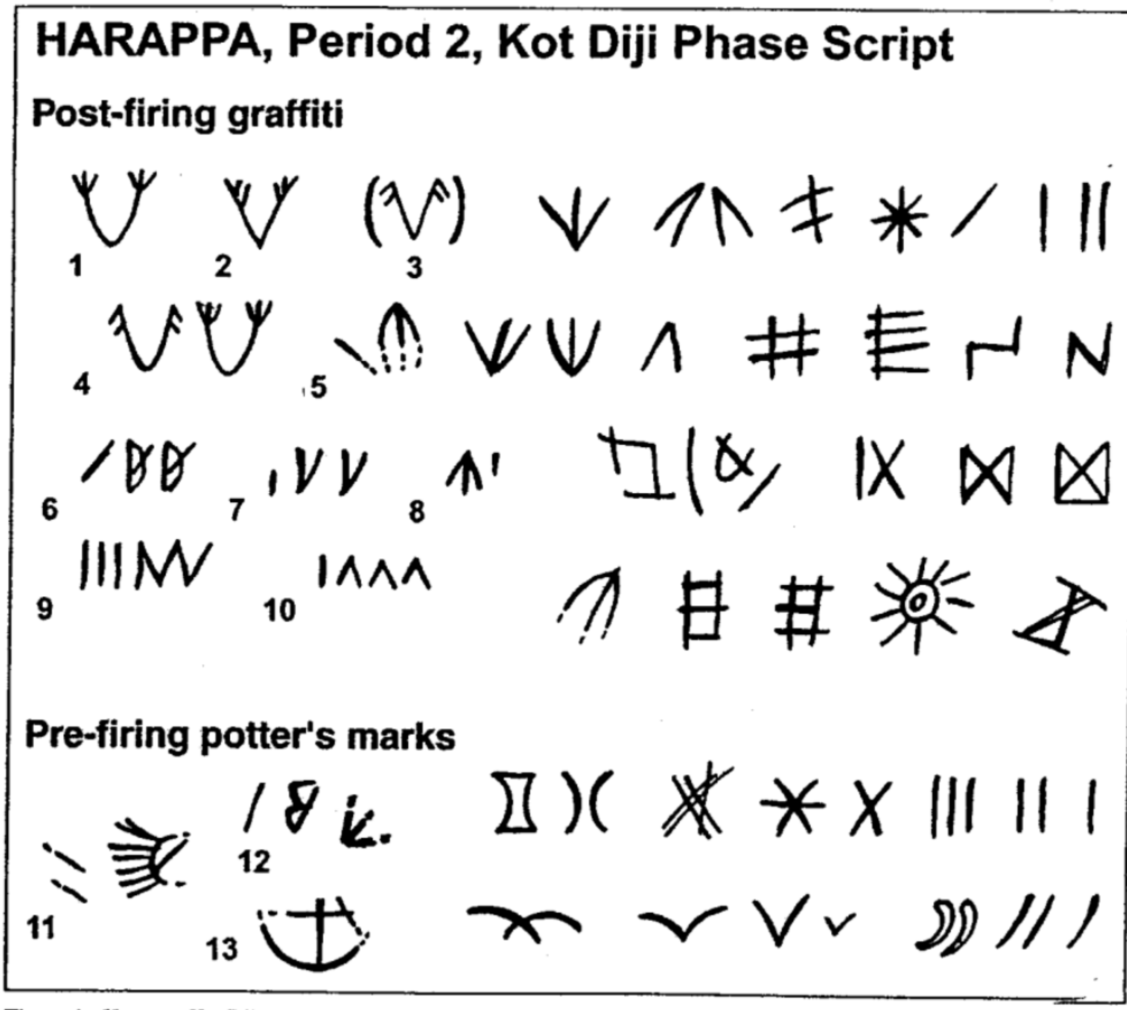

Figure 4. Harappa, Kot Diji Phase Script

We can note the beginning stages of creation of logo-syllables like: vēl-ūr, valai-ūr, vilampu, taṭṭi, pura, iru-vaṟi, etc.

In this Period-2, the sun is portrayed now not as a star ✳ as in Period-1, but as , which eventually became a hollow ellipse ○ in the mature stage of Indus script.

Phase 3: The Mature phase

The next stage is a great leap forward to a new punctuated equilibrium [3], with the blossoming of a new proto-Dravidian language and it's logo-syllabic vocabulary of more than 400 signs.

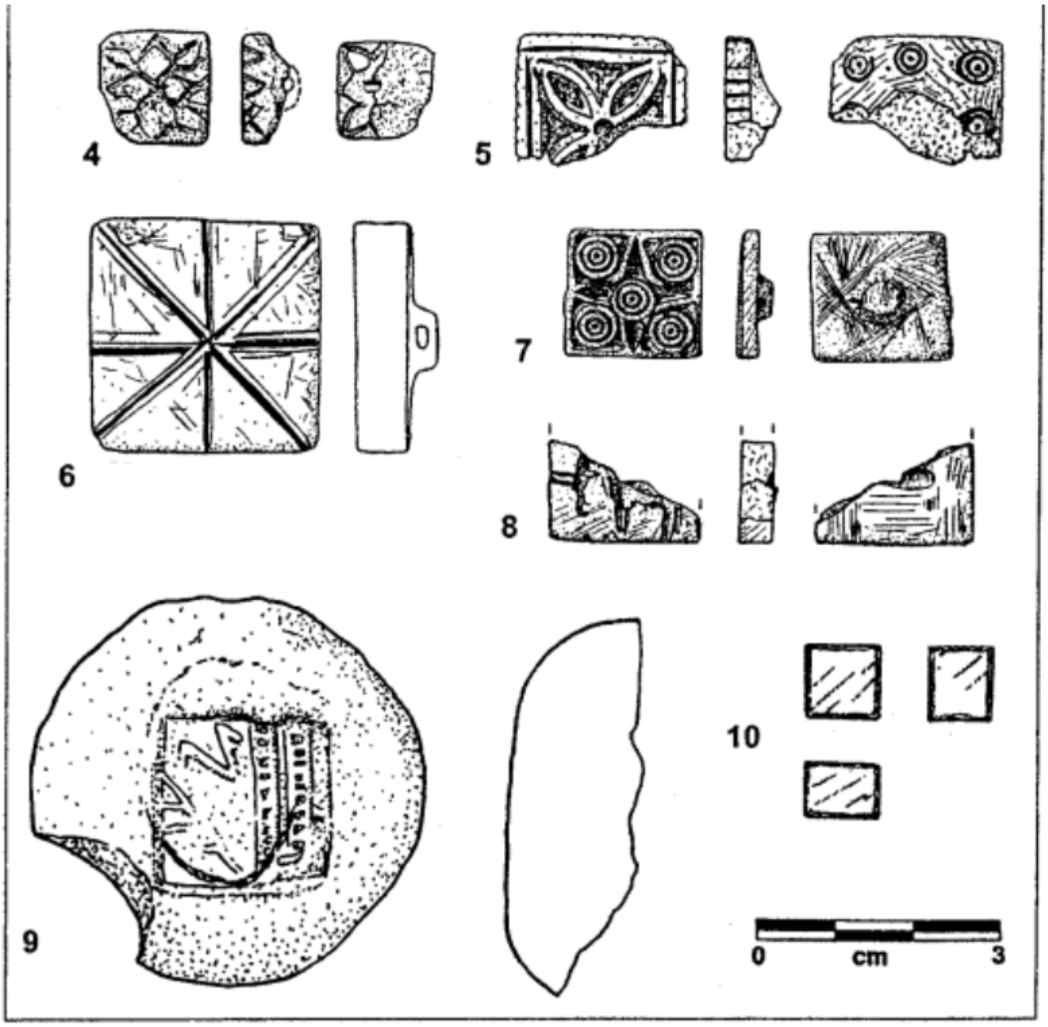

Figure 5. Harappa, Ravi and Kot Diji Phase Seals

1. Button Seal, bone, Ravi Phase Period 1 (H98-3503/8514-07)
2. Button Seal, glazed steatite, Kot Diji Phase, Period 2 (H96-2740/7469-01)
3. Button Seal, glazed steatite, Kot Diji Phase, Period 2 (H96/7458-01)

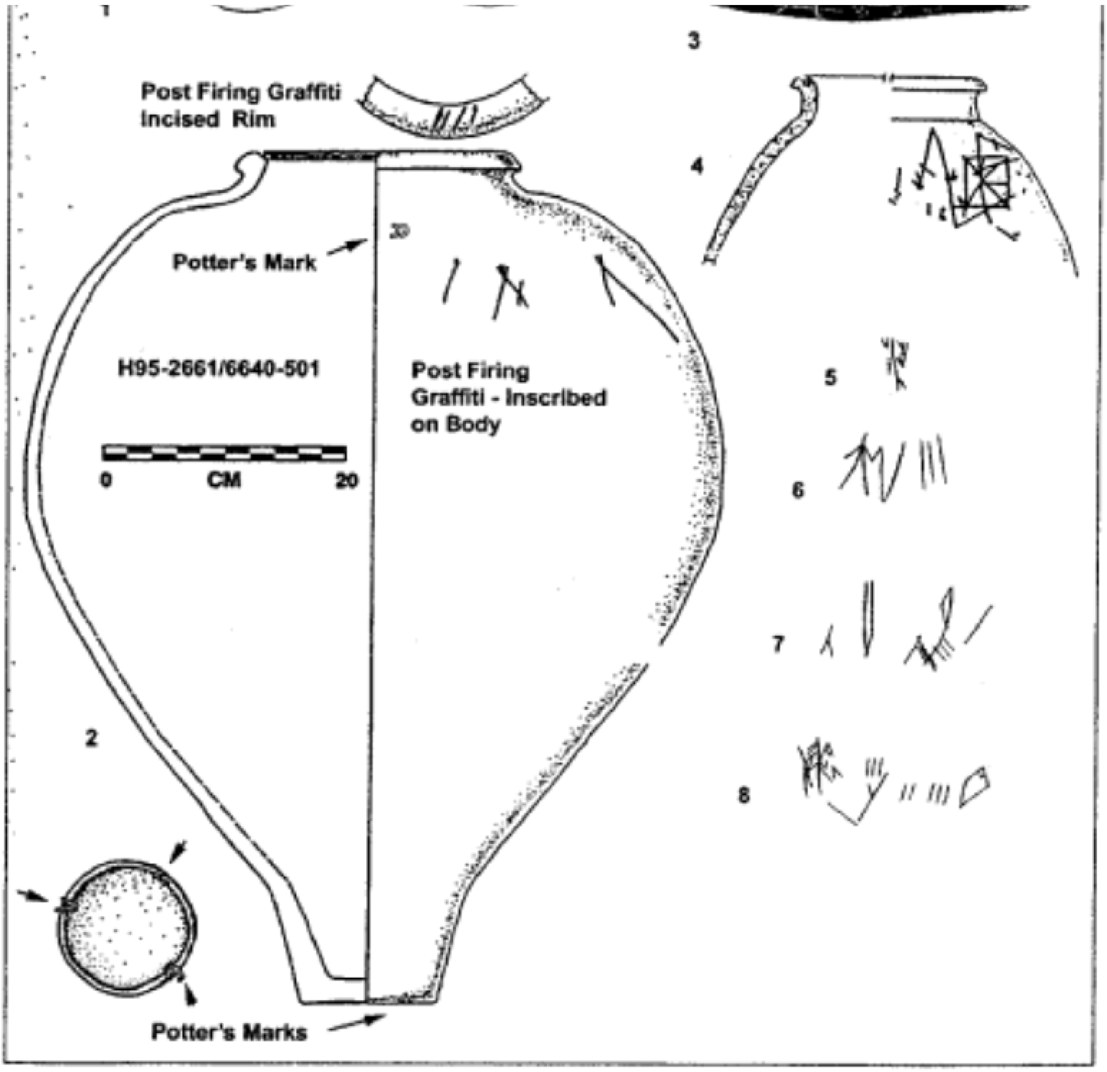

Figure 7. Harappa Period Inscribed Pottery

1. Black Slipped Jar, post-firing graffiti on rim and body (H95 / 5684-02)
2. Black Slipped Jar, post-firing graffiti on rim and body, pre-firing potter's marks on base and body (H95-2662 / 6640-501)
3. Large Red Slipped Black Band Jar, post-firing graffiti on body (H94 / 4290-14)
4. Plain globular jar, post-firing graffiti on body (H95 / 5630-15)

Early Food Producing Era	*circa + 7000 to 5000 B.C.*
Regionalization Era	*circa 5000 to 2600 B.C.*
Harappa Period 1A/B (Hakra/Ravi)	>3700-2800 B.C.
Harappa Period 2 (Kot Dijian)	2800-2600 B.C.
Integration Era	***2600 to 1900 B.C.***
Harappa Period 3A	2600-2450 B.C.
Harappa Period 3B	2450-2200 B.C.
Harappa Period 3C	2200-1900 B.C.
Localization Era	*1900 to 1300 B.C.*
Harappa Period 4 (Late Harappan)	1900-1800 B.C.
Harappa Period 5 (Late Harappan/Cemetery H)	1800-<1700 B.C.

Table 2. General Chronology: Indus Tradition and Harappa Chronology

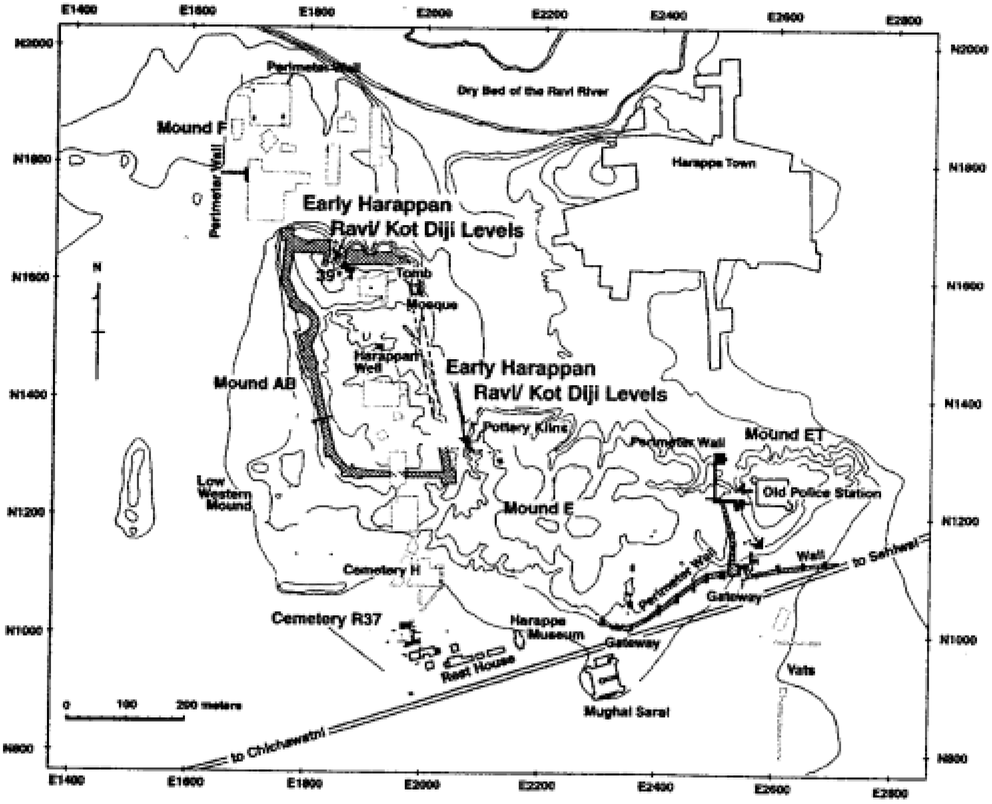

Figure 2. Map of Harappa

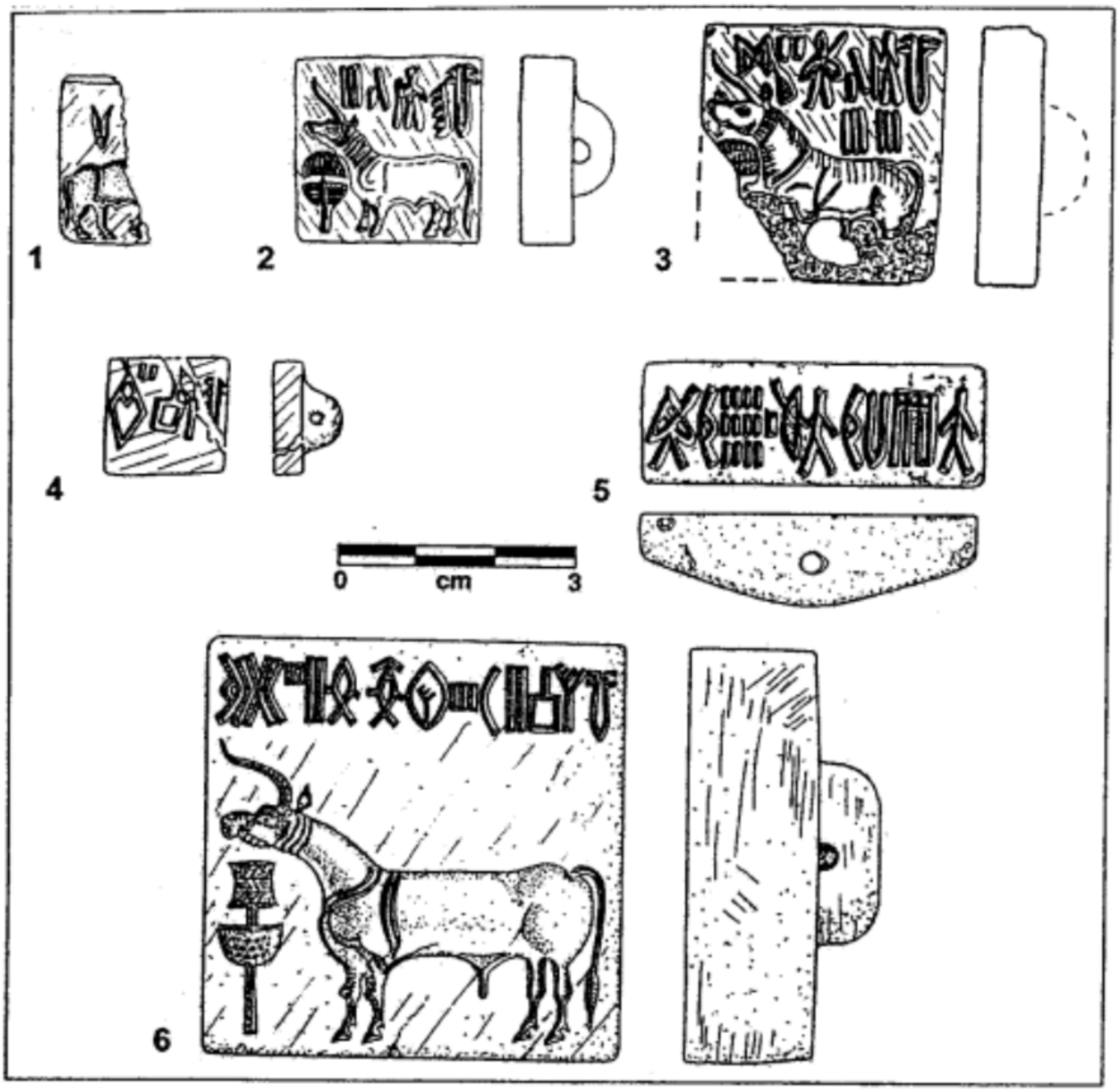

Figure 6. Harappa Phase Intaglio Seals

1. Steatite seal, Period 3A, (H90-1600/3166-01)
2. Steatite seal, Period 3B, (H95-2491/4690-01)
3. Steatite seal, Period 3B, (H95-2410 / 5145-55)
4. Steatite seal, Period 3B/3C, (H93-2170 / 4114-09)
5. Steatite seal, Period 3C, (H98-3491/8322-21)
6. Steatite seal, Period 3C, (H99-4064/8796-01)

Conclusion

Through these archaeological findings and the decipherment (Venkatesan, 2025) we can hypothesize that Indus polity was a class society, an advancement over the less productive clan-based societies. It is indeed a anomaly that unlike other Bronze-age societies it did not develop a strong religious ruling class and the chieftains did not grow into powerful kings as in Egypt, Sumeria and Elam.

From the decipherment it is clear that there are no personal names for officials or towns, only names derived out of professions. This cannot be just an accident of history. There must be social forces that did not permit such a lopsided growth. We point out one such process that could have prevented exaggerated concentration of wealth and inequality: the high velocity of barter trade across the Indus river and it's effect on social transformation. There is also constant messaging that calls for social intercourse using the Swastika symbol (which is deciphered in Proto-Dravidian as cerukoḷaḷ, meaning "joining-together/joining-in-barter"). Explicit calls are made for mass marriage between the provinces and towns, arranged by the collective state. There seems to be a conscious effort at increasing the velocity of exchange across provinces and towns, helping to create a homogenous society and along with it the proto-Dravidian language that emerged through the symbolism of trade and the barter system encoded by the Indus script (Venkatesan, 2025).

The massive network of canals and river-based transport, further aided by trade winds, provided a rapid means of physical transport of people and goods. This also aided in the creation of a homogenous society and polity. However, this complex homegeneous society can also breakdown quickly if any of these enabling factors also get bottlenecked. Without a strong centralized authority and large accumulation of wealth, a failure of trade winds or the drying of the river can cause a complete breakdown, leading to a rapid fragmentation and decline of the civilization.

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